

Spatial Analysis of Colorado Housing Prices Using Zillow Listing Data

by

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SECTION I: INTRODUCTION

Housing prices are strongly connected to geography. The value of a residential property is shaped by physical characteristics, such as square footage, number of bedrooms, number of bathrooms, and home type, but location also plays a major role. Homes are affected by local housing demand, employment centers, school districts, transportation access, neighborhood conditions, and regional market patterns. For this reason, housing data are especially appropriate for spatial analysis.

Spatial statistics provides a framework for studying these geographic patterns. In ordinary statistical analysis, observations are often treated as independent records in a table. In spatial analysis, the location of each observation is also considered. Homes that are near one another may share similar market conditions and local amenities, so nearby listings may show similar prices or cluster around major population centers.

The purpose of this project is to apply spatial statistics and mapping methods to Colorado Zillow housing listings. The dataset includes sale price, home type, city, ZIP code, beds, baths, square footage, latitude, and longitude. Because each listing includes geographic coordinates, each home can be treated as a spatial point. Colorado state and county polygon boundaries are used to provide geographic context.

The project has three main goals: to map Zillow listings within the Colorado boundary, to use regression analysis to examine how housing price relates to property characteristics and location, and to create an interactive Leaflet map for exploring individual listings. Together, these steps show how spatial data analysis provides insight beyond a table alone.

SECTION II: DATA DESCRIPTION

The dataset used in this project is the Colorado Zillow listing dataset. Each row represents one residential housing listing in Colorado. The response variable is listing price, while key predictors include beds, baths, area in square feet, home type, days on Zillow, latitude, and longitude. City and ZIP code are used to describe where listings are located.

Before analysis, the dataset was imported into R and cleaned. Duplicate listing IDs were removed, and records were filtered to keep observations with valid price, coordinate, bed, bath, and square footage values. Coordinate values were checked to ensure the points fell within a reasonable Colorado range, since incorrect coordinates would create misleading maps and distance calculations.

Several additional variables were created. Price per square foot was calculated to compare homes of different sizes, and log price was created because listing prices were strongly right-skewed. Price categories were also used to describe the distribution of listings.

Table 1. Overall Summary Statistics

Metric	Value
Number of listings	123
Number of cities	44
Number of ZIP codes	82
Minimum price	\$49,999
Median price	\$635,000
Mean price	\$3,972,322
Maximum price	\$300,000,000
Median area square feet	2,540
Median price per square foot	\$254

Table 1 summarizes the cleaned dataset. The sample contains 123 listings across 44 cities and 82 ZIP codes. The median price is \$635,000, while the mean price is approximately \$3.97 million. The large difference between the median and mean reflects the influence of a small number of extremely high-value listings. The median home size is 2,540 square feet, and the median price per square foot is approximately \$254.

SECTION III: EXPLORATORY SUMMARY STATISTICS

Before mapping and modeling, summary statistics were used to understand the structure of the housing listings. These summaries identified which cities appeared most often, how prices varied across places, and whether certain home types dominated the sample.

City-level summaries show that Colorado Springs and Denver had the largest number of listings. Colorado Springs had the highest listing count, while Denver had a higher median listing price among the cities with many listings. The top ten cities are shown in Table 2, and the full city-level table is included in Appendix A.

Table 2. Top 10 Cities by Number of Zillow Listings

City	Listing Count	Median Price	Mean Price	Median Price per Sq Ft
Colorado Springs	18	\$502,500	\$556,244	\$232
Denver	16	\$887,450	\$1,106,556	\$409
Littleton	8	\$700,000	\$857,125	\$245
Lakewood	6	\$637,500	\$652,475	\$262
Arvada	5	\$725,000	\$967,000	\$287
Centennial	5	\$650,000	\$670,800	\$235
Boulder	4	\$3,050,000	\$5,562,500	\$717
Castle Rock	4	\$667,500	\$652,225	\$212
Fort Collins	4	\$610,000	\$644,500	\$342
Aurora	4	\$515,000	\$501,250	\$211

Table 2 shows that the sample is concentrated around major Colorado housing markets, especially cities along the Front Range. Boulder has fewer listings than Colorado Springs or Denver, but it has a much higher median price, reflecting high-value listings in that area.

Table 3. Home Type Summary Statistics

Home Type	Listing Count	Median Price	Mean Price	Median Area Sq Ft
Single Family	115	\$650,000	\$4,215,370	2,637
Condo	4	\$432,000	\$402,250	944
Townhouse	4	\$555,000	\$554,750	1,560

Table 3 shows that the dataset is dominated by single-family homes. Condos and townhouses each appear only four times, so those categories should be interpreted cautiously. Because single-family homes make up most of the data, overall price patterns are mainly driven by that group.

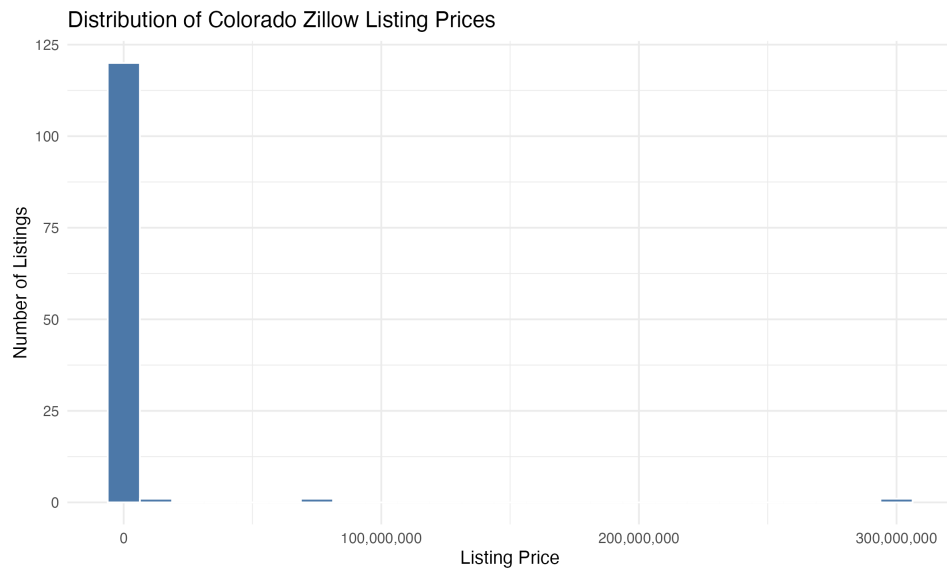


Figure 1. Distribution of Colorado Zillow listing prices.

The histogram shows a strongly right-skewed price distribution. Most listings fall in moderate price ranges, while a small number of luxury properties have extremely high prices. This supports the decision to use log price in the regression model.

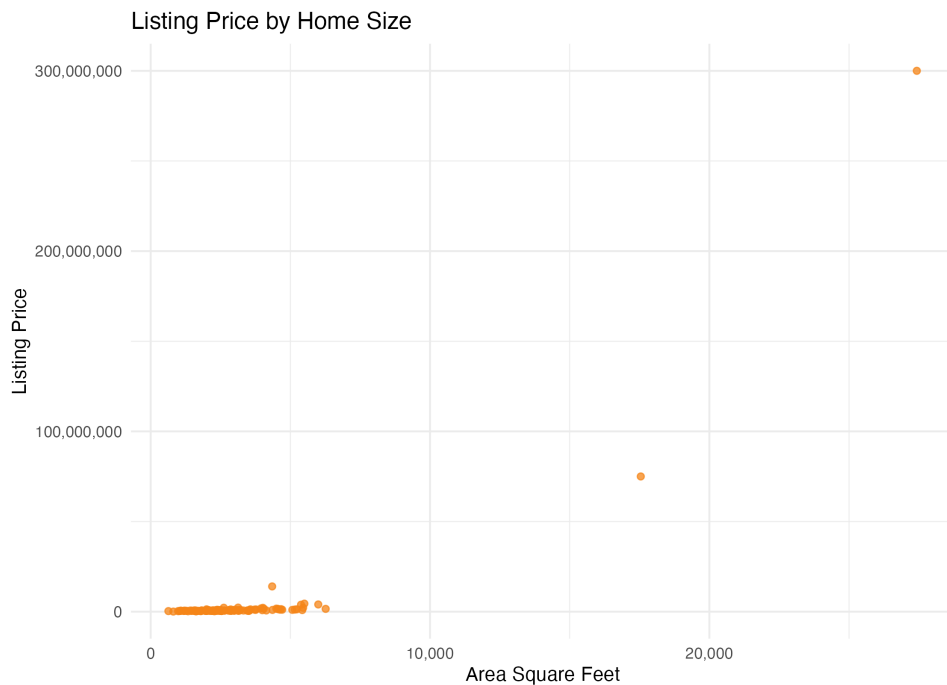


Figure 2. Relationship between listing price and area square feet.

The scatterplot shows that larger homes tend to have higher listing prices, but the relationship is not perfectly linear. Some homes have high prices because of location or luxury-market features, not only because of size.

SECTION IV: SPATIAL DATA PREPARATION

The Zillow listings were converted from a regular data frame into a spatial object using the `sf` package in R. Longitude and latitude were used as coordinates, and the coordinate reference system was set to WGS 84. After this conversion, each listing became a spatial point.

Polygon data were added for geographic context. Colorado state and county boundaries were obtained using the `tigris` package, which downloads U.S. Census boundary files. The state boundary shows the shape of Colorado, while county polygons show internal geographic divisions.

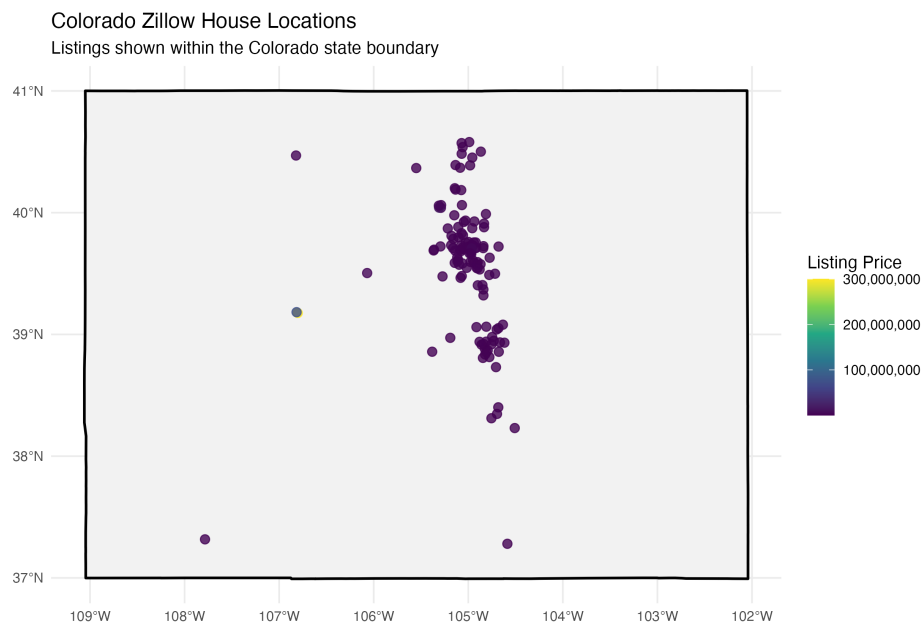


Figure 3. Zillow listing points within the Colorado state boundary.

SECTION V: STATIC MAP OF ZILLOW LISTINGS

The first major map displays Zillow listing points inside the Colorado boundary. The listings are not evenly distributed across the state; they are concentrated along the Front Range near Denver, Colorado Springs, Boulder, and surrounding communities.

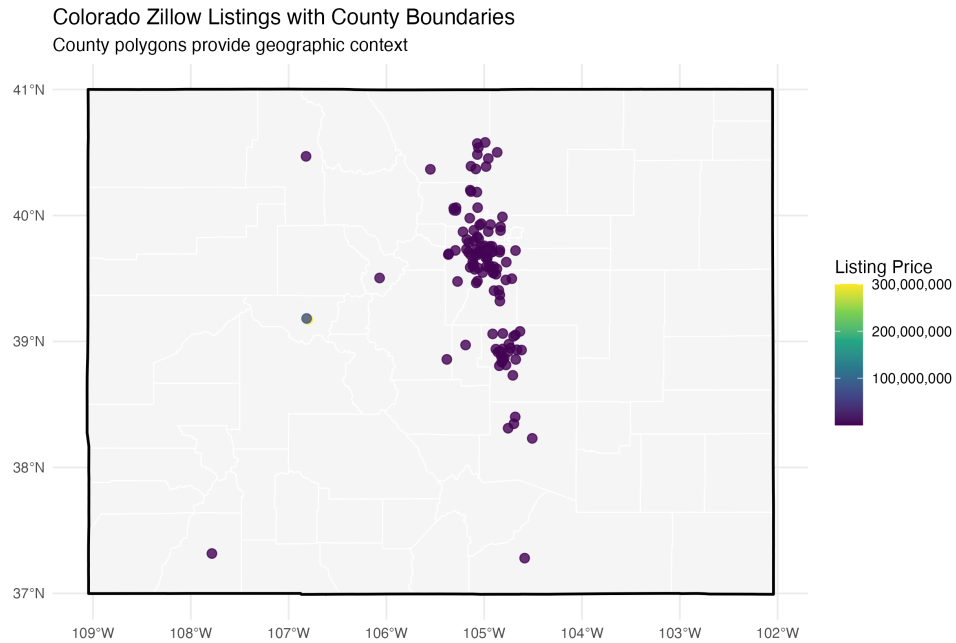


Figure 4. Zillow listing points with Colorado county boundaries.

Adding county boundaries provides more spatial context and makes it easier to identify the concentration of listings in urbanized and suburban areas. A smaller number of listings appear in mountain or western areas.

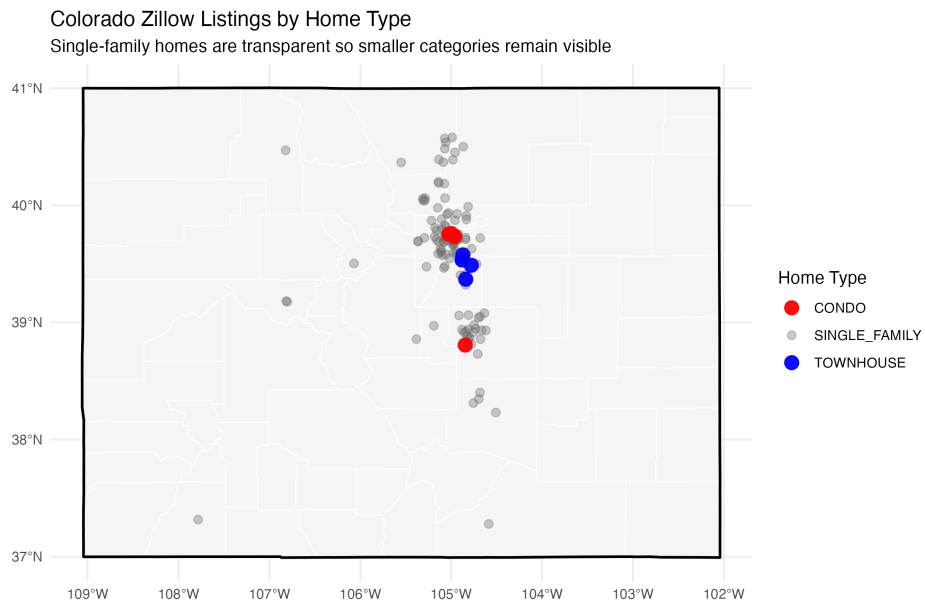


Figure 5. Zillow listings by home type.

The home type map shows that most listings are single-family homes. Single-family points are shown with transparency so that the smaller condo and townhouse categories remain visible.

SECTION VI: REGRESSION ANALYSIS

A regression model was fitted to examine how listing price relates to housing characteristics and location. The response variable was log listing price because the original price variable was strongly right-skewed. Predictors included beds, baths, area square feet, home type, days on Zillow, latitude, and longitude.

Table 4. Regression Coefficient Summary

Variable	Estimate	Std. Error	t Value	p-value
Intercept	-40.087	11.070	-3.621	0.0004
Beds	-0.020	0.048	-0.420	0.6751
Baths	0.111	0.054	2.045	0.0432
Area Sq Ft	0.000195	0.000040	4.880	<0.001
Single Family	0.442	0.222	1.993	0.0486
Townhouse	0.435	0.300	1.450	0.1498
Days on Zillow	-0.00477	0.00120	-3.984	0.0001
Latitude	0.163	0.071	2.295	0.0236
Longitude	-0.437	0.105	-4.165	<0.001

Table 4 shows that area square feet, baths, single-family home type, days on Zillow, latitude, and longitude were meaningful predictors in this model. Area square feet and bathrooms had positive relationships with log price. Days on Zillow had a negative coefficient, suggesting that homes listed longer tended to have lower predicted prices, holding other variables constant. The significant latitude and longitude coefficients support the importance of geography in housing price variation.

SECTION VII: REGRESSION RESIDUAL MAP

Residuals were calculated as actual listing price minus predicted price. A positive residual means the actual price was higher than the model prediction, while a negative residual means the actual price was lower. Mapping residuals helps show where the model overpredicts or underpredicts prices geographically.

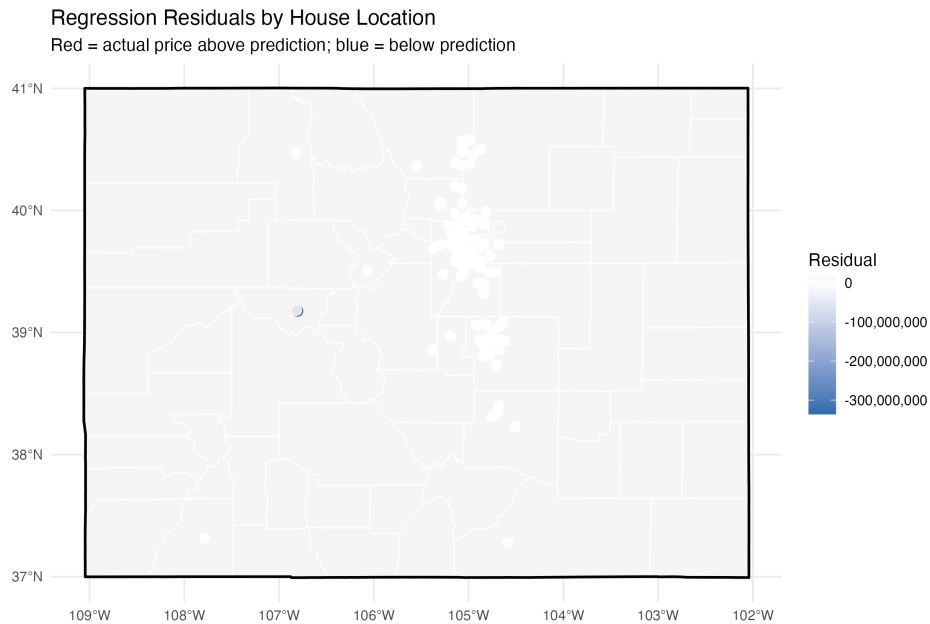


Figure 6. Regression residuals by house location.

The residual map connects the regression model back to geography. Large residuals may reflect local features not included in the model, such as neighborhood quality, views, school zones, luxury amenities, or other market factors.

SECTION VIII: NEAREST-NEIGHBOR SPATIAL POINT PATTERN ANALYSIS

A nearest-neighbor analysis was used to evaluate whether Zillow listings were randomly distributed, clustered, or dispersed. The method compares the observed mean distance from each point to its nearest neighbor with the expected distance under complete spatial randomness.

Table 5. Nearest-Neighbor Summary

Metric	Value
Number of listings	123
Study area convex hull square km	81,832.085
Listing intensity per square km	0.001503
Observed mean nearest-neighbor distance km	8.356
Expected mean nearest-neighbor distance km under CSR	12.897
Nearest-neighbor index	0.648
Point pattern interpretation	Clustered

The observed mean nearest-neighbor distance was 8.356 kilometers, compared with an expected distance of 12.897 kilometers under complete spatial randomness. The nearest-neighbor index was 0.648, which is less than 1. This indicates that the listings are spatially clustered.

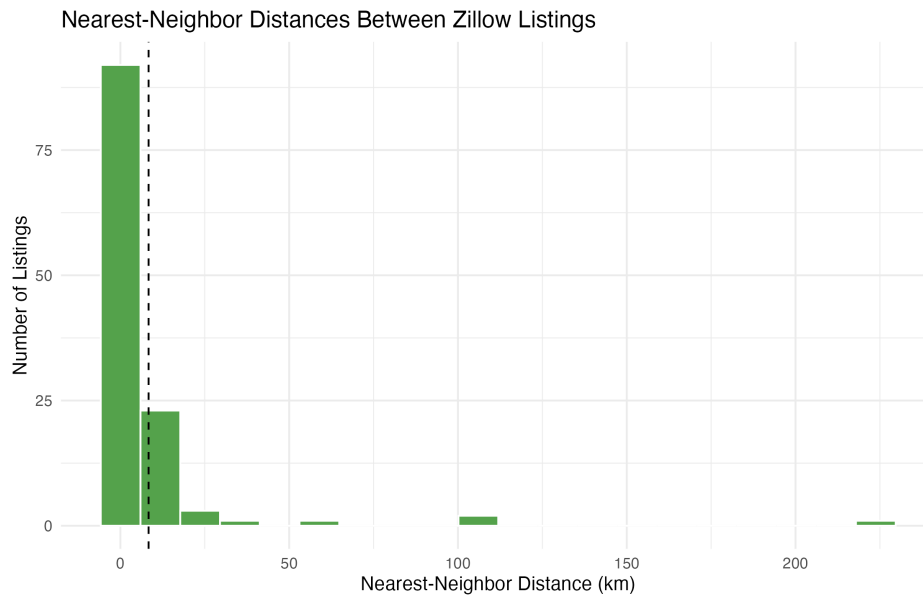


Figure 7. Distribution of nearest-neighbor distances.

SECTION IX: SPATIAL CLUSTER ANALYSIS

A simple spatial cluster analysis grouped listings based on their coordinates. This was exploratory and was used to summarize regional groups rather than make formal predictions.

Table 6. Spatial Cluster Summary

Cluster	Listing Count	Median Price	Mean Price	Median Area Sq Ft	Approximate Location
I	29	\$460,000	\$558,110	2,456	Southern Colorado / Colorado Springs area
II	3	\$75,000,000	\$125,241,633	17,544	Mountain / luxury-market listings
III	72	\$662,500	\$883,048	2,583	Denver and central Front Range area
IV	19	\$650,000	\$1,742,421	2,637	Northern Front Range / Boulder-area listings

The largest cluster is centered around Denver and the central Front Range. Another cluster captures southern Colorado, including Colorado Springs. A small luxury-market cluster contains only three listings but has extremely high mean and median prices, so it should be interpreted cautiously.

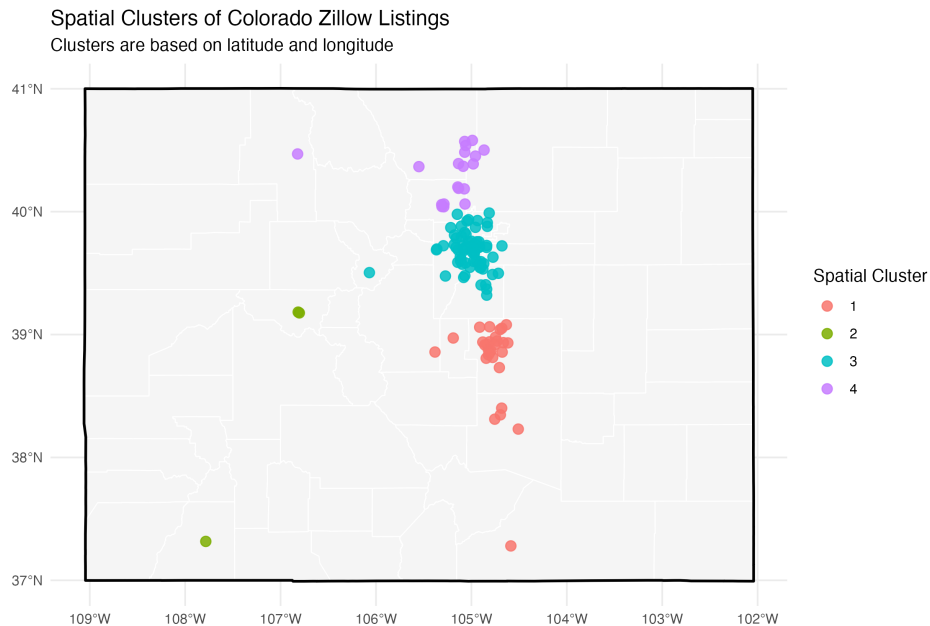


Figure 8. Spatial clusters of Colorado Zillow listings.

SECTION X: INTERACTIVE LEAFLET MAP

The final product is an interactive Leaflet map. Unlike a static map, the Leaflet map allows viewers to click individual points and inspect listing-level information, including address, city, actual price, predicted price, residual, home type, beds, baths, square footage, and spatial cluster.

Interactive map file: colorado_zillow_complete_interactive_map.html

The interactive map is useful because it combines the Colorado boundary, county polygons, listing points, price information, and model results into one exploratory tool.

SECTION XI: CONCLUSION

This project demonstrates how spatial statistics can be applied to housing data. Zillow listings were treated as spatial points using latitude and longitude, and Colorado polygon boundaries were added to provide geographic context. Summary statistics, maps, regression modeling, residual mapping, nearest-neighbor analysis, clustering, and an interactive map were used to examine the data.

The main finding is that the Zillow listings are spatially clustered rather than randomly distributed. The nearest-neighbor index of 0.648 indicates that listings are closer together than expected under complete spatial randomness. This agrees with the maps, which show many listings concentrated along the Front Range.

The regression model also showed that both property characteristics and location are related to listing price. Overall, the project shows that spatial analysis adds important value to real estate data because maps reveal where patterns occur, while tables and regression describe the patterns numerically.

APPENDIX A: FULL CITY-LEVEL SUMMARY STATISTICS

Appendix Table A1. Full City-Level Summary Statistics

City	Listing Count	Median Price	Mean Price	Median Price per Sq Ft
Colorado Springs	18	\$502,500	\$556,244	\$232
Denver	16	\$887,450	\$1,106,556	\$409
Littleton	8	\$700,000	\$857,125	\$245
Lakewood	6	\$637,500	\$652,475	\$262
Arvada	5	\$725,000	\$967,000	\$287
Centennial	5	\$650,000	\$670,800	\$235
Boulder	4	\$3,050,000	\$5,562,500	\$717
Castle Rock	4	\$667,500	\$652,225	\$212
Fort Collins	4	\$610,000	\$644,500	\$342
Aurora	4	\$515,000	\$501,250	\$211
Longmont	3	\$620,000	\$563,000	\$226
Pueblo West	3	\$260,000	\$285,967	\$142
Aspen	2	\$187,500,000	\$187,500,000	\$7,607
Evergreen	2	\$1,647,500	\$1,647,500	\$392
Monument	2	\$1,334,950	\$1,334,950	\$321
Englewood	2	\$967,500	\$967,500	\$308
Highlands Ranch	2	\$885,000	\$885,000	\$229
Lone Tree	2	\$752,000	\$752,000	\$359
Parker	2	\$721,500	\$721,500	\$264
Broomfield	2	\$564,950	\$564,950	\$245
Golden	2	\$557,500	\$557,500	\$301
Loveland	2	\$522,500	\$522,500	\$261
Thornton	2	\$502,000	\$502,000	\$269
Breckenridge	1	\$4,450,000	\$4,450,000	\$809
Steamboat Springs	1	\$2,175,000	\$2,175,000	\$831
Windsor	1	\$1,179,000	\$1,179,000	\$228
GOLDEN	1	\$1,095,000	\$1,095,000	\$363
Louisville	1	\$950,000	\$950,000	\$393
Peyton	1	\$890,000	\$890,000	\$164
Durango	1	\$724,900	\$724,900	\$326
Commerce City	1	\$675,000	\$675,000	\$163
Erie	1	\$650,000	\$650,000	\$233
Henderson	1	\$569,000	\$569,000	\$162
Johnstown	1	\$545,000	\$545,000	\$207
Westminster	1	\$525,000	\$525,000	\$426
Estes Park	1	\$500,000	\$500,000	\$419
Severance	1	\$495,000	\$495,000	\$173
Divide	1	\$450,000	\$450,000	\$183

Pueblo	1	\$449,000	\$449,000	\$193
Trinidad	1	\$449,000	\$449,000	\$179
Conifer	1	\$440,000	\$440,000	\$342
Brighton	1	\$399,900	\$399,900	\$114
Florissant	1	\$228,000	\$228,000	\$171
Yoder	1	\$179,000	\$179,000	\$110

APPENDIX B: R CODE

```
# STA6709 Spatial Statistics Semester Project
# Spatial Analysis of Colorado Housing Prices Using Zillow Listing
Data
# Xiaoling Sundberg
# May 03, 2026

# Step 1: Install packages
library(readr)
library(dplyr)
library(ggplot2)
library(sf)
library(tigris)
library(leaflet)
library(scales)

options(tigris_use_cache = TRUE)

# Step 2: Folder paths
project_dir <- "/Users/xiaolingsundberg/Desktop/STA6709/
SemesterProject"
data_dir <- file.path(project_dir, "DATA")
writeup_dir <- file.path(project_dir, "WRITE UP")

# Step 3: Read in data
zillow <- read_csv(
  file.path(data_dir, "colorado_zillow.csv"),
  show_col_types = FALSE
)

head(zillow)
dim(zillow)
names(zillow)

# Step 4: Clean data and keep only latitude, longitude, price,
beds, baths,
# and square footage. Also creates price per square foot and log
price.
```

```

zillow_clean <- zillow %>%
  distinct(zpid, .keep_all = TRUE) %>%
  filter(
    !is.na(latitude),
    !is.na(longitude),
    !is.na(price),
    !is.na(beds),
    !is.na(baths),
    !is.na(area_sqft),
    latitude >= 36.5,
    latitude <= 41.5,
    longitude >= -109.5,
    longitude <= -101.5
  ) %>%
  mutate(
    price_per_sqft = price / area_sqft,
    log_price = log(price),
    price_category = case_when(
      price < 500000 ~ "Below $500K",
      price < 750000 ~ "$500K-$749K",
      price < 1000000 ~ "$750K-$999K",
      TRUE ~ "$1M+"
    )
  )

head(zillow_clean)
dim(zillow_clean)

# Step 5: Summary statistics overall
overall_summary <- tibble(
  Metric = c(
    "Number of listings",
    "Number of cities",
    "Number of ZIP codes",
    "Minimum price",
    "Median price",
    "Mean price",
    "Maximum price",
    "Median area square feet",
    "Median price per square foot"
  ),

```

```

Value = c(
  nrow(zillow_clean),
  n_distinct(zillow_clean$city),
  n_distinct(zillow_clean$zipcode),
  min(zillow_clean$price),
  median(zillow_clean$price),
  mean(zillow_clean$price),
  max(zillow_clean$price),
  median(zillow_clean$area_sqft),
  median(zillow_clean$price_per_sqft)
)
)

overall_summary

# Step 6: Summary statistics by city
city_summary <- zillow_clean %>%
  group_by(city) %>%
  summarize(
    listing_count = n(),
    median_price = median(price),
    mean_price = mean(price),
    median_price_per_sqft = median(price_per_sqft),
    .groups = "drop"
  ) %>%
  arrange(desc(listing_count), desc(median_price))

head(city_summary, 10)

# Step 7: Summary statistics by home type
home_type_summary <- zillow_clean %>%
  group_by(home_type) %>%
  summarize(
    listing_count = n(),
    median_price = median(price),
    mean_price = mean(price),
    median_area_sqft = median(area_sqft),
    .groups = "drop"
  ) %>%
  arrange(desc(listing_count))

home_type_summary

```

```

# Step 8: Get Colorado polygon data from tigris (state and county)
us_states <- states(cb = TRUE, year = 2023)

colorado_boundary <- us_states %>%
  filter(STUSPS == "CO") %>%
  st_transform(crs = 4326)

colorado_counties <- counties(state = "CO", cb = TRUE, year = 2023)
%>%
  st_transform(crs = 4326)

plot(st_geometry(colorado_boundary), main = "Colorado State
Boundary")
plot(st_geometry(colorado_counties), main = "Colorado County
Boundaries")

# Step 9: Convert Zillow listings into spatial points (sf object)
zillow_sf <- st_as_sf(
  zillow_clean,
  coords = c("longitude", "latitude"),
  crs = 4326,
  remove = FALSE
)

zillow_sf
plot(st_geometry(zillow_sf), main = "Zillow Listings as Spatial
Points")

# Step 10: Map house locations inside Colorado state polygon
map_house_locations <- ggplot() +
  geom_sf(data = colorado_boundary, fill = "gray95",
    color = "black", linewidth = 0.8) +
  geom_sf(data = zillow_sf, aes(color = price),
    size = 2.6, alpha = 0.80) +
  scale_color_viridis_c(labels = comma) +
  labs(
    title = "Colorado Zillow House Locations",
    subtitle = "Zillow listings shown inside the Colorado state
boundary",
    color = "Listing Price"
  ) +

```

```

theme_minimal()

map_house_locations

# Step 11: Map house locations with county boundaries
map_house_locations_counties <- ggplot() +
  geom_sf(data = colorado_counties, fill = "gray96",
          color = "white", linewidth = 0.35) +
  geom_sf(data = colorado_boundary, fill = NA,
          color = "black", linewidth = 0.8) +
  geom_sf(data = zillow_sf, aes(color = price), size = 2.5, alpha =
0.80) +
  scale_color_viridis_c(labels = comma) +
  labs(title = "Colorado Zillow Listings with County Boundaries",
        subtitle = "County polygons provide geographic context for
house locations",
        color = "Listing Price") +
  theme_minimal()

map_house_locations_counties

# Step 12: Map houses by home type
map_home_type <- ggplot() +
  geom_sf(data = colorado_counties, fill = "gray96",
          color = "white", linewidth = 0.35) +
  geom_sf(data = colorado_boundary, fill = NA,
          color = "black", linewidth = 0.8) +
  geom_sf(data = zillow_sf, aes(color = home_type),
          size = 2.5, alpha = 0.85) +
  labs(title = "Colorado Zillow Listings by Home Type",
        subtitle = "Listings are mapped as spatial points over
Colorado county polygons",
        color = "Home Type") +
  theme_minimal()

map_home_type

# Step 13: Histogram for house price
price_histogram <- ggplot(zillow_clean, aes(x = price)) +
  geom_histogram(bins = 25, fill = "#4c78a8", color = "white") +
  scale_x_continuous(labels = comma) +
  labs(

```

```

    title = "Distribution of Colorado Zillow Listing Prices",
    x = "Listing Price",
    y = "Number of Listings"
  ) +
  theme_minimal()

price_histogram
# price is highly right skewed - need log transformation for
analysis

# Step 14: Scatter plot for price and square feet.
price_by_area <- ggplot(zillow_clean, aes(x = area_sqft, y =
price)) +
  geom_point(alpha = 0.75, color = "#f58518") +
  scale_x_continuous(labels = comma) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Listing Price by Home Size",
    x = "Area Square Feet",
    y = "Listing Price"
  ) +
  theme_minimal()

price_by_area

# Step 15: Fit a regression model
price_model <- lm(
  log_price ~ beds + baths + area_sqft + home_type + days_on_zillow
+
  latitude + longitude,
  data = zillow_clean
)

summary(price_model)

# Step 16: Show regression coefficients as a table
regression_coefficients <- as.data.frame(summary(price_model)
$coefficients)
regression_coefficients$Variable <-
rownames(regression_coefficients)

regression_coefficients <- regression_coefficients %>%

```

```

select(
  Variable,
  Estimate,
  `Std. Error`,
  `t value`,
  `Pr(>|t|)`
)

```

```

regression_coefficients

```

```

# Step 17: Add predicted prices and residuals
# Residual = actual price - predicted price.
# Positive residual means the house price is higher than predicted.
# Negative residual means the house price is lower than predicted.
zillow_model_results <- zillow_clean %>%
  mutate(
    predicted_log_price = predict(price_model, newdata =
zillow_clean),
    predicted_price = exp(predicted_log_price),
    residual = price - predicted_price,
    absolute_residual = abs(residual)
  )

```

```

zillow_model_results %>%
  select(address, city, price, predicted_price, residual) %>%
  head(10)

```

```

# Step 18: Map regression residuals to see location of over/under-
priced houses
zillow_model_sf <- st_as_sf(
  zillow_model_results,
  coords = c("longitude", "latitude"),
  crs = 4326,
  remove = FALSE
)

```

```

residual_map <- ggplot() +
  geom_sf(data = colorado_counties, fill = "gray96", color =
"white", linewidth = 0.35) +
  geom_sf(data = colorado_boundary, fill = NA, color = "black",
linewidth = 0.8) +

```

```

geom_sf(data = zillow_model_sf, aes(color = residual), size =
2.7, alpha = 0.85) +
  scale_color_gradient2(
    low = "#2b6cb0",
    mid = "white",
    high = "#c53030",
    midpoint = 0,
    labels = comma
  ) +
  labs(
    title = "Regression Residuals by House Location",
    subtitle = "Red = actual price above prediction; blue = actual
price below prediction",
    color = "Residual"
  ) +
  theme_minimal()

```

residual_map

```

# Step 19: Nearest-neighbor spatial analysis
# A nearest-neighbor index below 1 suggests clustering.
haversine_km <- function(lat1, lon1, lat2, lon2) {
  radius_km <- 6371.0088
  lat1 <- lat1 * pi / 180
  lon1 <- lon1 * pi / 180
  lat2 <- lat2 * pi / 180
  lon2 <- lon2 * pi / 180
  dlat <- lat2 - lat1
  dlon <- lon2 - lon1
  a <- sin(dlat / 2)^2 + cos(lat1) * cos(lat2) * sin(dlon / 2)^2
  2 * radius_km * asin(sqrt(a))
}

lonlat_to_xy_km <- function(data, lon_col = "longitude", lat_col =
"latitude") {
  lat0 <- mean(data[[lat_col]], na.rm = TRUE) * pi / 180
  x <- data[[lon_col]] * 111.320 * cos(lat0)
  y <- data[[lat_col]] * 110.574
  cbind(x, y)
}

coords <- zillow_clean %>% select(latitude, longitude)

```

```

n_points <- nrow(coords)
distance_matrix <- matrix(Inf, nrow = n_points, ncol = n_points)

for (i in seq_len(n_points)) {
  distance_matrix[i, ] <- haversine_km(
    coords$latitude[i],
    coords$longitude[i],
    coords$latitude,
    coords$longitude
  )
  distance_matrix[i, i] <- Inf
}

nearest_neighbor_distances <- apply(distance_matrix, 1, min)

xy_points <- lonlat_to_xy_km(zillow_clean)
hull_index <- chull(xy_points[, 1], xy_points[, 2])
hull_points <- xy_points[c(hull_index, hull_index[1]), ]

study_area_km2 <- abs(sum(
  hull_points[-1, 1] * hull_points[-nrow(hull_points), 2] -
  hull_points[-nrow(hull_points), 1] * hull_points[-1, 2]
) / 2)

listing_intensity <- n_points / study_area_km2
observed_mean_nn <- mean(nearest_neighbor_distances)
expected_mean_nn <- 1 / (2 * sqrt(listing_intensity))
nearest_neighbor_index <- observed_mean_nn / expected_mean_nn

point_pattern_interpretation <- case_when(
  nearest_neighbor_index < 0.9 ~ "clustered",
  nearest_neighbor_index > 1.1 ~ "more dispersed than random",
  TRUE ~ "approximately random"
)

nearest_neighbor_summary <- tibble(
  Metric = c(
    "Number of listings",
    "Study area convex hull square km",
    "Listing intensity per square km",
    "Observed mean nearest-neighbor distance km",
    "Expected mean nearest-neighbor distance km under CSR",
  )

```

```

    "Nearest-neighbor index",
    "Point pattern interpretation"
  ),
  Value = c(
    n_points,
    round(study_area_km2, 3),
    round(listing_intensity, 6),
    round(observed_mean_nn, 3),
    round(expected_mean_nn, 3),
    round(nearest_neighbor_index, 3),
    point_pattern_interpretation
  )
)

nearest_neighbor_summary

# Step 20: Histogram of nearest-neighbor distances
nearest_neighbor_histogram <- ggplot(
  tibble(nearest_neighbor_distance_km =
nearest_neighbor_distances),
  aes(x = nearest_neighbor_distance_km)
) +
  geom_histogram(bins = 20, fill = "#54a24b", color = "white") +
  geom_vline(xintercept = observed_mean_nn, linetype = "dashed") +
  labs(
    title = "Nearest-Neighbor Distances Between Zillow Listings",
    x = "Nearest-Neighbor Distance (km)",
    y = "Number of Listings"
  ) +
  theme_minimal()

nearest_neighbor_histogram

# Step 21: Simple spatial clusters
set.seed(42)
cluster_fit <- kmeans(xy_points, centers = 4, nstart = 20)

zillow_clustered <- zillow_model_results %>%
  mutate(spatial_cluster = cluster_fit$cluster)

cluster_summary <- zillow_clustered %>%
  group_by(spatial_cluster) %>%

```

```

summarize(
  listing_count = n(),
  median_price = median(price),
  mean_price = mean(price),
  median_area_sqft = median(area_sqft),
  latitude = mean(latitude),
  longitude = mean(longitude),
  .groups = "drop"
) %>%
arrange(spatial_cluster)

```

cluster_summary

```

# Step 22: Map spatial clusters
zillow_clustered_sf <- st_as_sf(
  zillow_clustered,
  coords = c("longitude", "latitude"),
  crs = 4326,
  remove = FALSE
)

```

```

cluster_map <- ggplot() +
  geom_sf(data = colorado_counties, fill = "gray96",
    color = "white", linewidth = 0.35) +
  geom_sf(data = colorado_boundary, fill = NA,
    color = "black", linewidth = 0.8) +
  geom_sf(data = zillow_clustered_sf, aes(color =
factor(spatial_cluster)),
    size = 2.7, alpha = 0.85) +
  labs(
    title = "Spatial Clusters of Colorado Zillow Listings",
    subtitle = "Clusters are based on latitude and longitude",
    color = "Spatial Cluster"
  ) +
  theme_minimal()

```

cluster_map

```

# Step 23: Interactive Leaflet map
zillow_clustered <- zillow_clustered %>%
  mutate(
    popup_text = paste0(

```

```

    "<b>Address:</b> ", address, "<br>",
    "<b>City:</b> ", city, "<br>",
    "<b>Actual Price:</b> $", comma(price), "<br>",
    "<b>Predicted Price:</b> $", comma(round(predicted_price,
0)), "<br>",
    "<b>Residual:</b> $", comma(round(residual, 0)), "<br>",
    "<b>Home Type:</b> ", home_type, "<br>",
    "<b>Beds/Baths:</b> ", beds, "/", baths, "<br>",
    "<b>Area:</b> ", comma(area_sqft), " sq ft<br>",
    "<b>Spatial Cluster:</b> ", spatial_cluster
  )
)

```

```

price_palette <- colorNumeric(
  palette = "viridis",
  domain = zillow_clustered$price
)

```

```

interactive_map <- leaflet() %>%
  addProviderTiles(providers$CartoDB.Positron) %>%
  addPolygons(
    data = colorado_counties,
    fillColor = "transparent",
    color = "gray60",
    weight = 0.6,
    opacity = 0.8,
    group = "Colorado counties"
  ) %>%
  addPolygons(
    data = colorado_boundary,
    fillColor = "transparent",
    color = "black",
    weight = 1.4,
    opacity = 1,
    group = "Colorado boundary"
  ) %>%
  addCircleMarkers(
    data = zillow_clustered,
    lng = ~longitude,
    lat = ~latitude,
    radius = ~pmax(4, pmin(18, sqrt(price) / 80)),
    color = ~price_palette(price),

```

```

    stroke = TRUE,
    weight = 1,
    fillOpacity = 0.75,
    popup = ~popup_text,
    group = "Zillow listings"
) %>%
addLegend(
  position = "bottomright",
  pal = price_palette,
  values = zillow_clustered$price,
  title = "Listing Price"
) %>%
addLayersControl(
  overlayGroups = c(
    "Colorado counties",
    "Colorado boundary",
    "Zillow listings"
  ),
  options = layersControlOptions(collapsed = FALSE)
)

```

interactive_map

```

# Step 24: Final results
cat("\nMain project results:\n")
cat("- Cleaned Zillow listings:", nrow(zillow_clean), "\n")
cat("- Median listing price: $",
comma(round(median(zillow_clean$price), 0)), "\n", sep = "")
cat("- Mean listing price: $",
comma(round(mean(zillow_clean$price), 0)), "\n", sep = "")
cat("- Nearest-neighbor index:", round(nearest_neighbor_index, 3),
"\n")
cat("- Point-pattern interpretation:",
point_pattern_interpretation, "\n")
cat("- Regression model uses log price as the response variable.
\n")
cat("- Interactive map object is named: interactive_map\n")

```